

MARS - Metrics and Abstractions for Reliable Agentic Systems

1 Summary

Agentic AI is moving into production faster than the community’s evaluation toolkit. Accuracy on a synthetic benchmark tells us whether an agent can finish a task. It tells us almost nothing about whether a team should deploy it, keep running it, or trust it once it is live. Production teams report that the things that decide those questions are cost, robustness to misuse, operational behavior under load, and the energy or carbon footprint of long-horizon reasoning. None of these have a shared measurement methodology. MARS is a one-day workshop on the new metrics and the reusable abstractions agentic AI needs once it is deployed. Cost across compute, memory, energy, dollars, and carbon is one worked example. Capability-per-watt ratios, robustness to prompt injection and agent misuse, and operational properties like tail latency at the plan level are others. The concrete deliverable is the first draft of an open, privacy-aware telemetry schema for production agents, scoped to hold the metrics arriving over the next year. MARS is organized by members of the team behind the MAP study of production agents and the MAST taxonomy of multi-agent failures, whose data is the clearest evidence that the current evaluation toolkit is too narrow for deployment.

2 Topic and Motivation

Recent capability benchmarks such as SWE-bench [10], GAIA [12], AgentBench [11], τ -bench [16], AppWorld [15], and SMDD-Bench [1] measure whether agents succeed in tool-using tasks with a long-horizon. They are almost entirely synthetic, and they reduce agent quality to a pass-or-fail score. Two studies looked at deployment from the outside. MAP surveyed practitioners across 306 respondents and 86 production agents [14], and the numbers are striking. Teams build agents for overwhelmingly economic ends, 80.3% for productivity and 72.7% to reduce human hours, yet 61.3% of deployed agents were never compared against any baseline, and 74.2% are evaluated by manual human inspection. Practitioners rank reliability, scalability, latency, and resource cost as their single largest challenge at 37.9%, ahead of every other category. MAST analyzed multi-agent traces and taxonomized where coordination breaks across fourteen failure modes [9]. Together, these numbers expose what the field has been slow to confront. Accuracy on a synthetic task does not measure most of what decides whether a deployed agent is useful. Cost is part of the gap. So is robustness when an agent set up for one task is induced into another. So is the efficiency of an agent’s reasoning per unit of energy, which recent intelligence-per-watt work has begun to formalize. So are operational properties teams report on, including scalability under multi-tenant batching, latency tails at the plan level, and sensitivity to prompt or input drift. The field is inventing the metrics that will replace pass-or-fail, and the work is scattered across venues that do not coordinate.

MARS brings the people inventing these metrics into one room, alongside the abstractions that let teams act on them. The workshop rests on two pillars.

2.1 Pillar I: New metrics for production agents

The first pillar concerns the metrics teams actually use, or wish they had, to decide whether an agent should be deployed and whether it should keep running. Several have emerged in the last year and more are still forming.

Cost. Recent work shows cost is not a scalar but a vector across compute, memory, energy, dollars, and carbon, each owned by a different stakeholder [5, 4, 7, 8]. Per-task, plan-granular attribution under multi-tenant serving is the missing piece, and methodology to recover honest per-task numbers is starting to appear.

Capability per unit of resource. Recent work proposes intelligence-per-watt and related ratios as a way to normalize agent capability against the energy or cost that produced it, exposing trade-offs the field has previously hidden inside aggregate accuracy scores. We expect ratios of this shape to enter standard reporting alongside accuracy.

Robustness to misuse and prompt injection. Once an agent acquires tools and operates over sensitive data, the question of whether it can be steered into unintended actions becomes a metric in its own right. Production teams are reporting incidents, and the measurement protocols are in flux.

Operational properties. Scalability under load, latency tails at the plan level, behavior under partial failure, and sensitivity to small prompt drift. These are everyday production concerns, and they have no shared measurement protocol yet. None of these metrics can be recovered from a practitioner survey or a text-trace taxonomy. All of these require instrumentation. The workshop’s contribution is to put their methodologies on one shared table and to push toward common reporting.

2.2 Pillar II: Abstractions and design for correct, efficient agents

Measurement without design is diagnosis without treatment. The second pillar concerns the abstractions that make agentic systems cheaper, faster, and safer by construction, and the question of which of them generalize and compose. Recent work includes plan-level rather than call-level instrumentation, which attributes cost to the unit a planner actually controls [4], plan-aware speculative serving that reads an agent’s declared plan as a prefetch oracle [3], multi-tier placement of inference across device, edge, and cloud [8, 2], and safety masks that bound an agent’s reachable action set so the agent operates inside a correctness certificate rather than around it [6]. The workshop treats these as the opening vocabulary of a design discipline, not finished answers, and asks the community to standardize the interfaces between them.

3 Why Now

The timing is specific. MAP and MAST both landed in 2025. Capability benchmarks proliferated across 2025 and 2026. Deployment is no longer hypothetical. MAP documents 86 production agents already in the field, and reports that teams judge success by whether an agent is good enough to replace a manual step rather than by a benchmark score. New metrics are arriving in the literature one paper at a time, scattered across venues that do not coordinate. There is no single venue where the people inventing them gather. Without this coordination, the field fragments along the metrics. Sustainability venues measure carbon, systems venues measure latency, security venues measure prompt-injection rates, and finance teams track dollars in private. None of these compose into a single picture of what a deployed agent costs or can be trusted to do. The abstractions that bound those costs keep being rebuilt per project. MARS holds the metric and abstraction agendas together in one place.

4 Workshop Scope

MARS’s scope is the practice of measuring and designing agentic AI for production deployment. The two pillars above set what the program accepts.

We invite work on metrics that go beyond accuracy. That spans cost across compute, memory, energy, dollars, and carbon, intelligence-per-watt and other capability-per-resource ratios, robustness to misuse and prompt injection, and operational metrics like scalability, tail latency at the plan level, and prompt-drift sensitivity. The infrastructure side spans telemetry schemas, abstractions for safety-bounded agency, plan-aware and multi-tier serving, measurement across the device-edge-cloud-satellite continuum, and open release of tooling.

Some areas the agent space at NeurIPS already covers well are out of MARS’s scope. Capability and reasoning benchmarks on synthetic tasks have their own venues. Single-model inference efficiency belongs to the foundation-model systems community. Workshops that use ML to improve systems components go in the opposite direction. Sustainability of training in isolation from agentic behavior has its own home. MARS does not duplicate any of those. Its focus is in-production measurement and abstractions, combined into a single discipline.

NeurIPS guidance rewards workshops that leave the community with more than slides. Alongside the standard paper track, MARS issues a call for anonymized production traces and hosts a working session whose output is a versioned draft of an open, privacy-aware telemetry schema, seeded on existing generative-AI observability conventions [13] and scoped to hold the metrics above. Privacy is treated as first-class, since MAP reports that 69.0% of production agents ingest confidential or sensitive data [14]. The workshop becomes the place where the measurement standard is born and gives attendees a reason to return in following years.

5 Call for Contributions

We solicit non-archival contributions, short and long, across topics chosen to be broad enough to draw submissions from communities the organizers do not personally lead.

- New metrics for production agents beyond accuracy, including capability-per-energy and capability-per-cost ratios.
- Robustness, misuse, and prompt-injection measurement for tool-using agents.
- Operational metrics including scalability, latency tails, and prompt-drift sensitivity at plan granularity.
- Instrumented measurement methodologies for long-horizon, in-production agentic workloads.
- Telemetry schemas and open data formats for agent traces, with privacy guarantees.
- Cross-stack cost attribution at task and plan granularity under multi-tenant serving.
- Energy and carbon accounting for multi-step reasoning and tool use.
- Abstractions for safety-bounded and constraint-masked agency.
- Plan-aware, speculative, and multi-tier serving for agentic inference.

- Measurement of agent workloads across the device, edge, cloud, and satellite continuum.
- Reproducibility and open release of measurement tooling and shared datasets.

6 Format, Schedule, and Interaction

MARS is a one-day, in-person workshop of roughly eight hours, weighted toward contributed work, posters, and structured working sessions rather than a wall of invited talks.

- Opening and framing by organizers, with a short live demonstration of a multi-metric production trace.
- Two morning invited talks, one from the practice-and-failure lineage, one from the measurement-and-systems lineage.
- A contributed-talk session of four to six selected papers.
- A poster session over an extended break, designed to surface in-progress measurement tooling.
- A short demo block, fifteen-minute hands-on demonstrations of open measurement tools submitted by contributors.
- The schema working session, the central community-building block, producing a versioned draft of the open trace schema, followed by a closing breakout to assign governance and next steps.
- Two moderated panels, one per pillar, both moderated by Negar Arabzadeh (tentative). **Panel 1, on metrics:** *When Benchmarks Don't Predict Production*, bringing benchmark authors, practitioners reporting deployment incidents, and industry agent owners onto why current evaluations miss the failure modes that matter at scale and what the new metric set should be. Anchored by the 2026 industry retrospectives documenting a 37 percent gap between lab benchmark scores and production performance. **Panel 2, on abstractions:** *What Production Agents Actually Need*, bringing researchers, industry deployers, and safety practitioners onto which agent design patterns actually compose in production, including verifier-based architectures, plan-level instrumentation, and bounded-autonomy interfaces.
- Two afternoon invited talks, one on abstractions for safe agency, one from an industry production deployment.

7 Invited Speakers

Three confirmed, four awaiting reply. Final slate will span seniority, gender, geography, and academic-industry balance.

- **Confirmed.** Hamed Haddadi (Imperial College London), measurement, privacy, on-device intelligence. Ali Shahin Shamsabadi (Brave Software), privacy-preserving ML and industry deployment. Sahar Abdelnabi (ELLIS Institute Tübingen), prompt injection and adversarial LLMs.
- **Tentative.** Azalia Mirhoseini (Stanford), scaling and serving for agentic reasoning. Niloofar Mireshghallah (CMU), privacy and contextual integrity. Ion Stoica (UC Berkeley), production systems for LLM serving.

8 Expected Attendance and Evidence of Demand

This is the first edition of MARS. We expect 120 to 150 attendees, drawn from machine-learning systems, sustainable computing, networking and edge systems, agent evaluation, and agent safety. MARS is the first venue to address these communities jointly, so attendance should exceed a single-community workshop.

9 Diversity and Inclusion

The organizing team, speaker slate, and program committee will span gender, career stage, geography, institution type, and the academic-industry divide, including researchers new to organizing alongside experienced ones. The call invites first-time submitters with light mentoring. The multi-metric framing is inclusive by design, since each metric area brings its own constituency.

10 Logistics

Location preference. Sydney, then Paris. Sydney is the primary venue, with Paris listed as the secondary option for European attendees and speakers.

Concurrent proposals. None of the organizers are submitting a related workshop proposal to NeurIPS 2026 or to another venue.

Submissions, reviewing, and conflicts. All contributions are non-archival via OpenReview and must present new or in-progress work; previously published or NeurIPS-accepted work is out of scope. We will size the PC for three reviews per paper, meet the mandatory September notification, post accepted talk titles publicly, and bar same-institution review or submissions by organizers and their students. No special technical needs.

References

- [1] SMDD-Bench Authors. Smdd-bench: Can llm agents do small-molecule drug design under oracle budgets? *arXiv preprint arXiv:2605.21740*, 2026.
- [2] Hannaneh Barahouei Pasandi et al. Orbitmesh: A four-tier inference serving system across the direct-to-cell satellite edge. In *IEEE/ACM Symposium on Edge Computing (SEC)*, 2026.
- [3] Hannaneh Barahouei Pasandi et al. Palse: Plan-aware prefetch for agentic llm serving at the edge. In *IEEE/ACM Symposium on Edge Computing (SEC)*, 2026.
- [4] Hannaneh Barahouei Pasandi et al. Plan-level carbon attribution for agentic ai workloads. In *Under review*, 2026.
- [5] Hannaneh Barahouei Pasandi et al. The reasoning tax: Profiling test-time compute carbon in agentic ai workloads. In *5th Workshop on Sustainable Computer Systems (HotCarbon)*, 2026.
- [6] Hannaneh Barahouei Pasandi et al. Safe wireless mac protocol synthesis with an interference epidemic threshold and bounded reasoning drift response. In *IEEE International Conference on Network Protocols (ICNP)*, 2026.
- [7] Hannaneh Barahouei Pasandi et al. The single-call fallacy: Why carbon-aware scheduling misses agentic ai’s real drivers. In *5th Workshop on Sustainable Computer Systems (HotCarbon)*, 2026.
- [8] Hannaneh Barahouei Pasandi et al. Skycascade: Carbon-aware cascading for mobile llm serving across device, edge, and cloud. In *5th Workshop on Sustainable Computer Systems (HotCarbon)*, 2026.
- [9] Mert Cemri, Melissa Z Pan, Shuyi Yang, Lakshya A Agrawal, Bhavya Chopra, Rishabh Tiwari, Kurt Keutzer, Aditya Parameswaran, Dan Klein, Kannan Ramchandran, et al. Why do multi-agent llm systems fail? volume 38, 2026.
- [10] Carlos E. Jimenez, John Yang, et al. Swe-bench: Can language models resolve real-world github issues? In *International Conference on Learning Representations (ICLR)*, 2024.
- [11] Xiao Liu, Hao Yu, et al. Agentbench: Evaluating llms as agents. *International Conference on Learning Representations (ICLR)*, 2024.
- [12] Grégoire Mialon, Clémentine Fourrier, et al. Gaia: A benchmark for general ai assistants. In *International Conference on Learning Representations (ICLR)*, 2024.
- [13] OpenTelemetry Authors. Opentelemetry semantic conventions for generative ai systems, 2025. OpenTelemetry Specification.
- [14] Melissa Z Pan, Negar Arabzadeh, Riccardo Cogo, Yuxuan Zhu, Alexander Xiong, Lakshya A Agrawal, Huanzhi Mao, Emma Shen, Sid Pallerla, Liana Patel, et al. Measuring agents in production. 2025.
- [15] Harsh Trivedi et al. Appworld: A controllable world of apps and people for benchmarking interactive coding agents. In *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2024.
- [16] Shunyu Yao, Noah Shinn, Pedram Razavi, and Karthik Narasimhan. τ -bench: A benchmark for tool-agent-user interaction in real-world domains. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

Organizer Information

Lead organizers (co-leads). Hannah B. Pasandi, University of California, Berkeley. h.pasandi@berkeley.edu Hannah is a networking and on-device intelligence researcher whose work treats hand-tuned system components as learnable modules governed by closed-form or empirically bounded correctness constraints, a program she terms self-driving systems. Her recent agentic-AI work spans both workshop pillars, including per-task energy and carbon attribution for agentic reasoning [5, 4, 7], carbon-aware multi-tier serving [8], plan-aware speculative serving at the edge [3], four-tier inference across the satellite edge [2], and constraint-masked agentic synthesis under a safety certificate [6]. She holds a CoNEXT 2025 Best Paper and Best Contribution Award and was named an N2Women Rising Star. She has co-chaired four ACM workshops at MobiSys, MobiCom, and CoNEXT between 2020 and 2021 (including the ACM CoNEXT’21 main workshop track), served as Publicity Chair for IEEE PerCom SmartEdge’21 and IEEE ICNP HDR-NETs’20, and moderated or organized panels at ACM CoNEXT’21 and SIGCOMM’21. Her diversity work includes organizing the N2Women workshop at CoNEXT’21 and ongoing mentorship of PhD and graduate students across NC A&T, UC Santa Barbara, Old Dominion University, and the University of Virginia.

Jon Saad-Falcon, Stanford University. [email: jonsaadfalcon@stanford.edu](mailto:jonsaadfalcon@stanford.edu) Jon is a PhD researcher at the Scaling Intelligence Lab at Stanford, working on scaling and evaluating large language model systems for multi-step reasoning and agentic workflows. His recent work spans efficient inference, evaluation methodology for agent capabilities, and the production-deployment frontier of compound AI systems. He has contributed to program committees and events at the Scaling Intelligence Lab.

Co-organizers. The workshop is co-organized by an 8-person team spanning UC Berkeley, Stanford, Barcelona Supercomputing Center, Université Grenoble Alpes, and Virginia Commonwealth University. Each one-line bio below names the perspective the organizer brings to the program.

- **Negar Arabzadeh**, UC Berkeley (Co-organizer, confirmed). Postdoctoral researcher at Berkeley working on evaluation and retrieval for large-scale language-model systems; co-author on the MAP study of production agents. [email: negara@berkeley.edu](mailto:negara@berkeley.edu)
- **Melissa Pan**, UC Berkeley (Organizer, confirmed). Berkeley researcher and lead author on MAP and co-author on MAST, working on practitioner-grounded measurement of deployed agentic AI. [email: melissapan@berkeley.edu](mailto:melissapan@berkeley.edu)
- **Sina Darabi**, Barcelona Supercomputing Center (Organizer, confirmed). BSC researcher on systems and infrastructure for large-scale AI workloads, bringing the European supercomputing-center perspective on production deployment. [email: sinad1367@gmail.com](mailto:sinad1367@gmail.com)
- **Franck Rousseau**, Université Grenoble Alpes and Grenoble INP-Ensimag, LIG lab (Organizer, confirmed). Associate professor working on networking, wireless and sensor systems, mobility, and security with applications to smart-home and smart-city deployments. [email: franck.rousseau@imag.fr](mailto:franck.rousseau@imag.fr)
- **Tamer Nadeem**, Virginia Commonwealth University (Organizer, confirmed). Professor of Computer Science and Associate Director of the VCU Cybersecurity Center, leader of the MuSIC lab, working on wireless networking, mobile computing, and intelligent transportation systems. [email: tnadeem@vcu.edu](mailto:tnadeem@vcu.edu)
- **Avanika Narayan**, Stanford Scaling Intelligence Lab (Organizer, tentative). Stanford researcher working on LLM systems and serving. [email: avanikan@stanford.edu](mailto:avanikan@stanford.edu)

The roster covers three continents, five institutions, and a mix of senior faculty, mid-career researchers, and PhD students. MAP and MAST are represented through co-authors rather than as the parent papers we extend. Cybersecurity, privacy, networking, and supercomputing perspectives anchor the program alongside ML systems.

Program committee. The PC will be sized for three reviews per submission with no reviewer asked to handle more than three papers, targeting fifteen to twenty-five members depending on submission volume. The candidate pool is anchored by MAP and MAST co-authors not on the organizer list (approximately thirty researchers across UC Berkeley, IBM Research, Stanford, UTUC, and other institutions), the labs of confirmed speakers (Imperial, ELLIS Tübingen, Brave), and the Scaling Intelligence Lab, Sky Computing Lab, LIG Drakkar group, VCU MuSIC Lab, and BSC. Coverage spans ML systems, sustainable computing, networking, edge ML, agent safety, robustness, and privacy. Final committee with confirmed acceptances will be published with the call by August.

Organizing experience. The team combines senior workshop experience with junior-researcher leadership. Tamer Nadeem (VCU) has served on ACM and IEEE program and organizing committees across networking and security and on editorial boards. Franck Rousseau (Grenoble) has co-organized multiple workshops in the networking and IoT space. Melissa Pan and Negar Arabzadeh (Berkeley) coordinated the MAP and MAST programs across thirty-plus contributors. Jon Saad-Falcon and Avanika Narayan have contributed to Stanford Scaling Intelligence Lab events. Hannah B. Pasandi (lead) has co-chaired four ACM workshops at MobiSys, MobiCom, and CoNEXT, and served on

multiple program committees in mobile and edge systems.